BEYOND EVALUATION USE

An Empirical Study of Involvement and Use/Influence in Multi-Site Evaluations

A Survey of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education Field
Report 7

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Section 1: Introduction

Beyond Evaluation Use is a research project studying the use and influence of the evaluations of four National Science Foundation (NSF)-funded programs by examining the relationship between the extent of involvement of evaluation stakeholders and the long-term impact of the evaluations on project staff; the science, technology, engineering, and mathematics (STEM) education community; and the evaluation community. The project was funded by NSF in 2004 and conducted by a team of researchers at the University of Minnesota. This research project developed a series of reports and case studies that examine ideas related to evaluation use, involvement and influence.

This research project targeted the evaluations of the following four large-scale, multi-site NSF programs: Advanced Technological Education (ATE); Collaboratives for Excellence in Teacher Preparation (CETP); Local Systemic Change through Teacher Enhancement (LSC); and Math Science Partnership - Research, Evaluation and Technical Assistance project at Utah State University (MSP-RETA). The four targeted program evaluation projects differ in the extent to which the principal investigators (PIs) of the funded projects within each of the programs were actively involved in the development and implementation of the evaluations, with ATE being the most evaluator-driven and MSP-RETA being the most PI-driven.

While research exists on use and influence of primary intended users through the utilization-focused evaluation principles of Patton (2008), the current study looked at influence beyond those primary intended users by examining influence on secondary or unintended users. The primary goal of this research was to examine the broad impact of the four program evaluation projects on the field, rather than the uses and influence on NSF as the intended user. This examination included the following components: 1) an online survey of the field, 2) a survey of evaluation and STEM education journal editors, 3) surveys and follow-up interviews with NSF project PIs, and 4) a citation analysis. This

report discusses two of these components: the online field survey and the survey of journal editors, both of which examined the impact of the four program evaluation projects on the broader fields of STEM education and evaluation. The online s STEM education field survey was designed to assess the broad impact of the ATE, CETP, LSC, and MSP-RETA program evaluation projects on members of the STEM education and evaluation community by tracking the perceived influence of the evaluations, their PIs, and related publications. The remaining two components are described in separate reports (Greenseid, et al., 2007; Toal, et al., 2008).

The broad impact or influence of evaluations is a relatively new topic in the evaluation community, as demonstrated in recent theoretical articles published by Drs. Mel Mark, Gary Henry, Karen Kirkhart, and Marv Alkin (Henry & Mark, 2003; Kirkhart, 2000; Mark & Henry, 2004; Alkin & Taut, 2003), and empirical investigations conducted by Carol Weiss and Tina Christie (Christie, 2007; Weiss, Murphy-Graham, & Birkeland, 2005). Influence is of particular interest in large multi-site evaluations because of the substantial amount of public funds used to support both the projects and the related evaluation efforts. The notion of affecting the field is inherent in the dissemination requirements in the four targeted evaluations, as it is in most NSF grants.

The following questions guided this research:

1. To what extent have individuals in the fields of STEM education perceived the ATE, CETP, LSC, and MSP-RETA evaluations, their PIs, and related publications as influential?
2. What factors have contributed to the perceived influence of the evaluations, their PIs, and related publications?

Section 2: Methodology

Assumptions and hypotheses

The research team was interested in whether or not ATE, CETP, LSC, and MSP-RETA evaluations, their PIs, and related publications would appear among the names and titles of influential evaluation-related people, products, and activities reported by members of the STEM education and evaluation communities. If no such names or titles emerged, then the conclusion might be that the evaluation had little or no influence. However, if survey respondents mentioned the names and titles of the four targeted program evaluations, then this might indicate a presence of a network of influence among the four targeted programs and STEM educators and evaluators.

The research team reasoned that participants in one of the four targeted program evaluation projects would be more likely than non-participants to name that evaluation in response to the survey. The team hypothesized that high-participation evaluations would be more well-known and more widely disseminated by projects PIs. This knowledge and dissemination would, in turn, result in greater impact or influence on the STEM field. In essence, the team wanted to explore the idea that there might be two levels of impact: one that would have direct influence on the program or project staff and another that would show indirect influence on the broader field. As mentioned earlier, the researchers discuss the examination of the effect on projects in separate case study reports; and two of the methods used to examine the effect on the field are presented in this report.

Network analysis

Following a suggestion of the research advisory committee during its meeting in Minneapolis in October 2007, the research team considered the use of network analysis to track the influence of the four evaluations. A network is a web of one's relations with others in which the patterns of relationships affect behavior (Berkowitz, 1982) and access to resources, support, and information (Wellman & Frank, 2000). In December 2007, the team had a phone consultation with an expert in Social Network Analysis (SNA), Dr. Noshir Contractor, Professor and Director of the Science of Networks in

Communities (SONIC) research group at Northwestern University. Dr. Contractor confirmed the appropriateness of SNA for our research and provided suggestions on how to best implement the study. An online survey was selected as the primary data collection method to obtain egocentric social network data. In an egocentric social network, each person is part of a network centered on a root individual. Studies of egocentric or personal networks examine how an individual's ties enable resource flow. The application of network analysis techniques was intended to identify evaluation influence in the network of STEM educators based on the presence and centrality of the names of the four targeted program evaluation projects and evaluators among the survey responses.

The original expectation was to identify and analyze an egocentric social network, i.e., the network that was incoming to the four PIs of the targeted program evaluation projects, with the size of that network corresponding to the level of influence or the most frequently cited being considered the most influential. Then, based on demographic information about PIs who were responding to the study, the team would be able to determine if any patterns existed in the types of people nominating the evaluations. Ultimately, it would provide a sense of how broad-ranging the impact of that evaluation might have been.

Sampling frame

Reasoning that it was important to examine an informed group, even though this group might introduce bias that would skew the findings in favor of larger or more robust networks of influence, the research team selected PIs of active education-related NSF grants drawn from the grant database of the NSF Directorate for Education and Human Resources. A December 2007 search of the database of NSF grants awarded from April 1, 1997 to April 1, 2007 produced a total of 2896 awards. Because PIs were the population of interest, 314 recipients of graduate fellowships and graduate teaching fellowships were removed from the dataset. In addition, PIs with multiple awards were

included only once, yielding a final dataset of 2301 PIs. The grants were from various NSF programs, and the PIs represented various universities and research institutions from across the United States. From that dataset, a research team member used SPSS software to draw a random sample of 350 individuals for the online field survey.

Survey design

Pre-pilot surveys

The research team deliberated over survey items that would effectively capture the idea of evaluation influence and began compiling potential items based on personal expertise and relevant literature. In January 2008, a pre-pilot survey was sent to a random sample of 90 PIs from the NSF grantee database (separate from the sample of 350 drawn for the actual survey.) These PIs were divided into three subgroups; and each subgroup was sent a portion of the survey addressing one of the three potential sources of influence: (1) large scale, multi-site STEM education evaluations, (2) evaluation publications, or (3) evaluators.

The first group was asked: "What is the most influential large scale, multi-site STEM education evaluation that you are aware of?" They were asked to provide the name of an evaluation and evaluator or organization that conducted that evaluation, as well as the reasons they thought it was the most influential evaluation. The second group received the question "What STEM evaluation publication (report, instrument, article, website, etc.) do you think has been most influential?" and was prompted to indicate the author, title, and type of publication, as well as the reasons why they thought it was influential. Finally, the third group was asked to list one person whom they believed was the most influential evaluator in the field of STEM education and to explain why they deemed that person influential. The response rate for the pre-pilots was 41% (37/90). The qualitative analysis of the pre-pilot responses provided valuable insight that shaped the revision and development of items for the final survey.

Think alouds

As an important part of the survey development process, in March 2008 three external experts were asked to complete a draft of the online field survey and provide their reactions to the survey's content, format, and ease of understanding. In addition, through email or online-based think alouds, the experts were asked to provide their thoughts on how the survey items could be improved. Their suggestions were incorporated into the survey as deemed appropriate by the research team. In addition, each member of the research team participated by taking the survey and sharing her or his suggestions with the entire team. As a result, a number of changes were made to both the survey and the email invitation. Those changes were related to the wording of some survey questions and statements, as well as to the survey format.

Pilot survey

In the next step of survey development, in April 2008 the research team piloted the Web-based survey with another random sample of 30 individuals from the NSF grant PIs database. The response rate was 33% (10/30). The research team used the pilot survey process to check the functionality of the survey and data downloads. Based on the pilot group's responses, the survey was slightly modified. The survey wording and format were finalized by the end of April 2008.

Conducting the field survey

To design the online survey, the research team employed an online survey tool offered by the College of Education and Human Development (CEHD) at the University of Minnesota. The survey tool allowed for the delivery of web-based surveys, the collection of data from survey participants, and the export of data into usable formats for manipulation and analysis with software such as MS Excel or SPSS.

The finalized online survey was administered to a random sample of 350 individuals from the database of over 2000 active NSF grant PIs under Education and Human

Resource Directorate in the NSF database. The data were collected from April 29, 2008 through May 27, 2008. All PIs from the sample received an initial invitation email with a description of the survey, an explanation of its use, and the Human Subjects approval number and phone numbers to call with questions. The email also contained a link to the online field survey with a personalized login and password. The team sent reminders to non-respondents one and two weeks following the first email.

The survey was comprised of 20 multiple choice and open-ended items (see Appendix A). The first three questions asked the PIs if they were familiar with one or more influential, large scale, multi-site STEM education evaluations; influential publications prepared/disseminated by large scale, multi-site STEM education evaluations; and influential evaluators who have conducted large scale, multi-site STEM education evaluations. Based on their answers, the respondents were automatically triaged to the follow-up questions eliciting responses about perceptions of what specific evaluations, publications, and evaluators might have been influential on the field of STEM education. In addition, those who answered "yes" to one of the first three questions were prompted to select up to two answers from a randomized list of reasons for why they found the named evaluations, publications, or evaluators to be influential.

Of the 350 invitation emails, 3% (11/350) were returned as undeliverable. Those 11 inoperable email addresses were replaced with new, randomly sampled PIs who were also sent survey invitation emails. Of those invitations, there appeared to be 10 successful deliveries and 1 delivery failure. The final sampling frame was 349 – the number of NSF PIs with valid email addresses and no delivery failures. The overall response rate was 54% (191/349).

STEM education and evaluation journal editors' survey

In a further attempt to measure the influence of large scale multi-site program evaluations on the field of STEM education evaluation, the team added another

dimension to the study – a survey of journal editors was added to the online field survey described in this report.

Reasoning that peer-reviewed publications, like program publications, presentations, and personal networks, are a means of influencing the field, the research team set out to collect information about the awareness of and ability to name influential evaluations, publications, and evaluators among the editors of STEM education and evaluation journals.

The research team identified six peer-reviewed publications that represented the field of STEM education and evaluation. The thinking was that these publications would be likely to have received manuscripts or published articles on topics related to the four program evaluations of interest.

The journals whose editors were contacted are:

- American Journal of Evaluation (AJE),
- Evaluation and Program Planning,
- Journal for Research in Mathematics Education (JRME),
- Journal of Research in Science Teaching (JRST) Information,
- School Science and Mathematics Association Journal (SSMA), and
- Science Education

In October 2008, the chief editors of these journals were sent an email asking them to name an influential evaluation, evaluator, and article in their respective journals based on a large scale, multi-site STEM education program. In addition, the editors were also asked to identify their professional association affiliations, their profession, and whether they had been involved in any of the four programs of interest. After a reminder email and subsequent telephone call in mid-October, three of the six responded to the questions. The other three editors did not respond.

Section 3: Survey Findings

Demographic information

The following table shows basic demographic characteristics of the survey respondents. These characteristics aligned with our expectations of the group. One item of note in these characteristics is the number of NSF grants the respondents were involved in at the time of the survey. Given the high number of respondents involved in two or more grants at the time of the survey, researchers were cognizant that such an engaged group could bias the size and robustness of reported social network connections, thereby limiting the ability to generalize to the overall field of STEM education. The researchers sought to identify a network and felt that this group would have greater exposure to STEM education evaluation dissemination activities through conferences and NSF program meetings. As a result, this group would have greater awareness of influential evaluations, publications and evaluators, and thus increase the possibility of creating a bias in favor of greater levels of knowledge.

Table 1. Respondent Demographics

Demographic Characteristics of Respondents		
Demographic Characteristic		***Percent***
Gender		
	Male	56.5
	Female	41.4
	No response	2.1
Number of NSF Grants the respondent was involved with at the time of the survey		
	1 NSF grant	34.0
	2-5 NSF grants	47.1
	> 5 NSF grants	17.3
	No response	1.6
Professional Background of Respondent		
	STEM Education/Researcher	52.4
	Scientist/Engineer/Mathematician	30.4
	Non-STEM Education/Researcher	6.8
	Other (Curator, Cultural Anthropologist, Economic Historian, Media Producer, Publisher)	6.3
	STEM Evaluator	1.6
	No response provided	2.5

General findings

This section will highlight general findings regarding influence on the STEM education and evaluation fields and later, the findings related to the four program evaluation projects of interest will be discussed. The research team set out to examine a number of factors that might begin to untangle influence and to identify network patterns regarding the influence of evaluations, publications, and evaluators on the field. The team explored the possible relationships between a respondent's NSF program experience, professional association membership, demographic characteristics, and involvement in one of the four program evaluation projects and their reported knowledge of influential evaluations, publications, and evaluators.

The survey first asked respondents if they were familiar with one or more influential, large scale, multi-site STEM education evaluations. The responses indicated that 64 of the 191 respondents (34%) knew of at least one influential large scale, multi-site STEM education evaluation. Figure 1 shows responses to this first question.

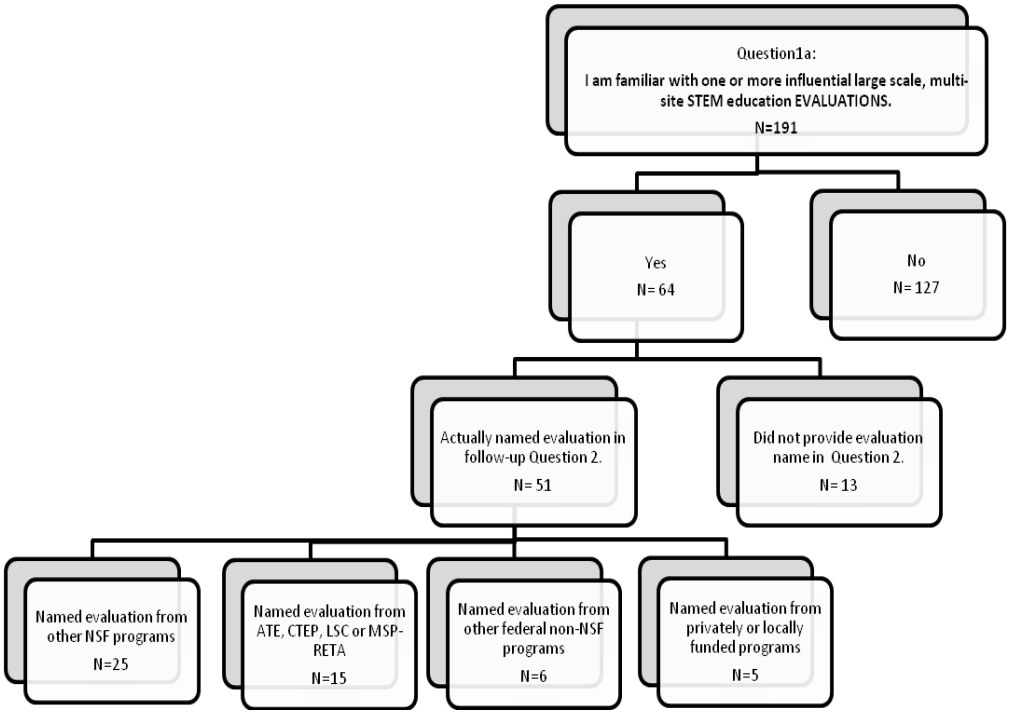


Figure 1 Item Responses for Question 1a - *I am familiar with one or more influential large scale, multi-site STEM education EVALUATIONS* **and Question 2 -** *Based on your experience, please name one influential large scale, multi-site STEM education program evaluation.*

Respondents were then asked if they were familiar with one or more influential publications produced by or resulting from a large-scale, multi-site STEM education evaluation. Responses showed that 57 of the 191 respondents (30%) knew of at least one such influential publication. Figure 2 shows responses to this question regarding influential publications.

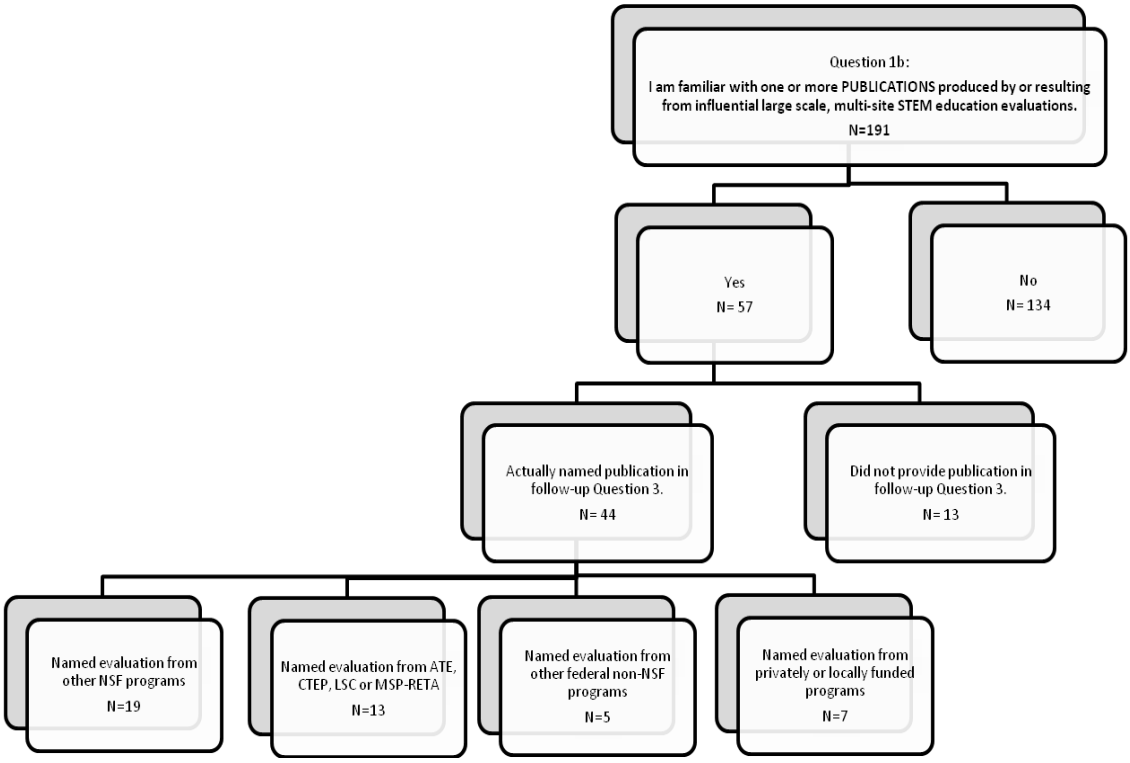


Figure 2 Item Responses for Question 1b - *I am familiar with one or more influential PUBLICATIONS produced by or resulting from large scale, multi-site STEM education evaluations* **and Question 3 -** *Based on your experience, please name one influential PUBLICATION prepared/disseminated by large scale, multi-site STEM education evaluations.*

Next, the survey asked respondents if they were familiar with one or more influential evaluators who conducted large scale, multi-site STEM education evaluations. Seventy of the 191 respondents (36%) indicated that they knew of one or more influential evaluators. Responses regarding familiarity with influential evaluators are noted in Figure 3. If a respondent answered "no" to each of these three questions, they were directed to five demographic questions and their survey was complete.

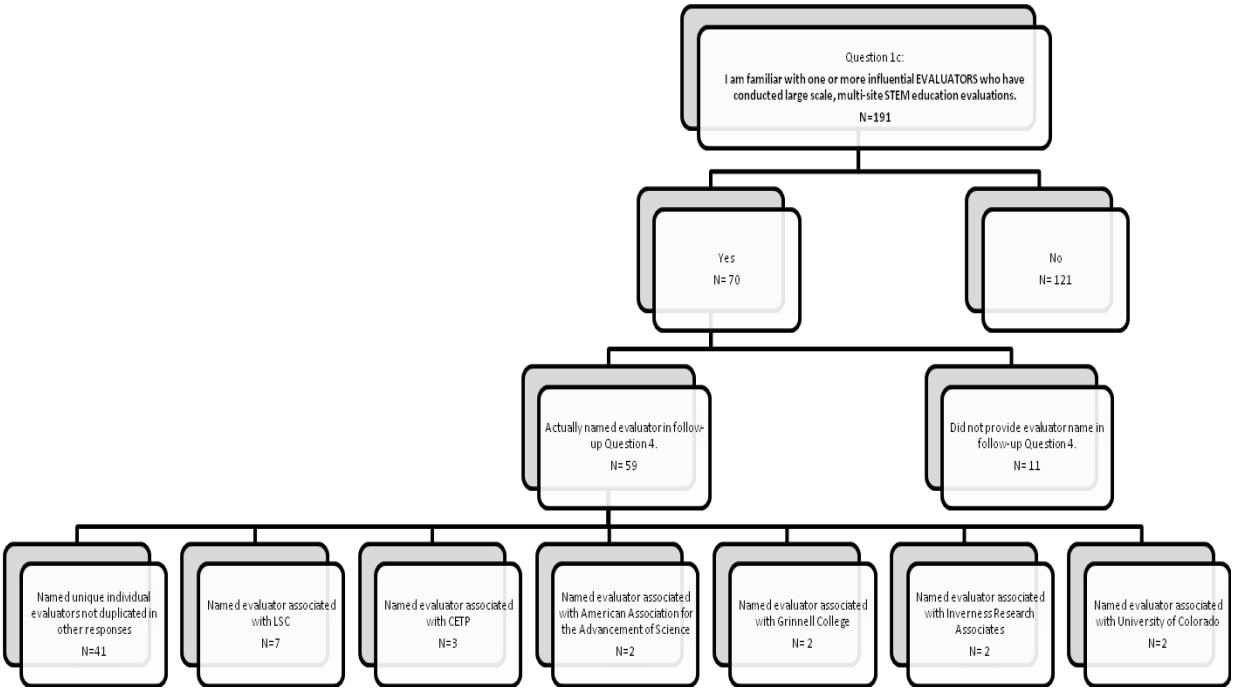


Figure 3 Item Responses for Question 1c - *I am familiar with one or more influential EVALUATORS who have conducted large scale, multi-site STEM education evaluations* **and Question 4 -** *Based on your experience, please name one influential evaluator in the field of STEM education. Please provide that person's full name and organizational affiliation.*

Respondents who answered "yes" to any of the first three questions continued with the survey and were asked a number of additional questions that would help investigate the presence of a social network of influence. Respondents who noted that they were aware of any influential large scale, multi-site STEM education evaluations, publications or evaluators were prompted to identify by name one or more of those evaluations, publications or evaluators. Figures 1, 2, and 3 also show that though respondents indicated familiarity with an influential evaluation, publication or evaluator, they did not necessarily provide one by name when prompted with related follow up questions.

The respondents were then asked to identify their reasons for naming an evaluation, publication, or evaluator as influential. This information was collected to help the research team to understand more about the nature of influence. Respondents were

asked to select from a randomized list of reasons for influence for each evaluation, publication, and evaluator. Respondents could select up to two reasons for influence for each category of evaluation, publication and evaluator. Given that, respondents provided 117 responses for reasons evaluations were influential, 96 responses regarding reasons publications were influential and 107 reasons an evaluator was influential. The selected reasons differed for influence of evaluations, publications, and evaluators. For instance, when citing the top reasons an evaluation was influential, respondents noted:

Reasons Cited for Evaluations' Influence *(n=117)*	***# of times***
It evaluated an important program	19
It was high quality work	15
It utilized sound methodology	14
I was involved in this evaluation	12
It presented both qualitative and quantitative findings	11
This is the only large scale, multi-site STEM education program evaluation of which I am aware	7
It offered important recommendations	7
It produced useful evaluation tools	6
It produced both positive and negative findings	6
Evaluation plans, instruments or reports have been widely disseminated	5
NSF recommended it to me	3
I learned about it in a workshop	3
It caused/contributed to important changes	3
It has had a far-reaching influence on both policy and practice	3
My colleagues have frequently used its products	1
I learned about it in a reputable publication	1
I was forced to use it	1

Examining the kinds of publications respondents mentioned showed that evaluation reports (n=20) were the most common followed by journals (n=13) and then books (n=8). Respondents cited the following as top reasons for publications being influential:

Reasons Cited for Publications' Influence *(n=96)*	***# of times***
It presented both qualitative and quantitative findings	14
It provided important recommendations	13
It was high quality work	12
It had far-reaching influence on both policy and practice	10
It utilized sound methodology	9
My colleagues have frequently used it	7
It evaluated an important program	7
It caused/contributed to important changes	6
It presented both positive and negative findings	6
It was widely disseminated	4
It presented useful evaluation tools	3

I heard about it from people I trust	2
NSF recommended it to me	1
I learned about it in a workshop	1
I learned about it in a reputable publication	1

Finally, when identifying reasons a particular evaluator was influential, respondents noted:

Reasons Cited for Evaluators' Influence *(n=107)*	***# of times***
Has demonstrated a deep understanding of STEM education	17
Has utilized sound evaluation/research methodology	11
Contributed significantly to the knowledge base of evaluation	11
Has had considerable experience in STEM evaluation	10
Evaluated an important program(s)	9
Has worked with me in the past	9
Provided important recommendations	8
Helped to strengthen STEM education	5
Was recommended by people I trust	5
Elevated the discussion about importance of program evaluation	5
Was recommended by NSF	4
Contributed to important changes	4
Disseminated his/her work (plans, instruments, reports, etc.) widely	3
Is a personal acquaintance of mine	2
Has been a proponent of building evaluation capacity	2
Is the only evaluator in the field of STEM education of which I am aware	1
Was written about in a reputable publication	1

The researchers conducted analyses looking at relationships between familiarity (with evaluations, publications, and evaluators) and several respondent characteristics, namely the number of NSF grants currently held, gender, and professional association membership. From this analysis, only one respondent characteristic showed a relationship of statistical significance. Specifically, the number of NSF grants held by the respondents was related to their familiarity with an evaluation and with an evaluator. Using SPSS 16 to explore cross-tabs and Chi square statistics, the researchers found a relationship between the number of NSF grants respondents currently held and their familiarity with at least one influential multi-site STEM education evaluation. There was a small statistically significant relationship between these with χ^2 (2, N= 188) = .208, p=.017. Researchers also found a small, but statistically significant relationship between

the number of NSF grants a respondent was involved with at the time of the survey and their familiarity with an influential evaluator of a multi-site STEM education evaluation. The relationship between these is indicated by χ^2 (2, N= 188) = .210, p=.016. Given the relatively low number of respondents in this analysis, it is important to consider these findings with caution.

Findings related to ATE, CETP, LSC and MSP-RETA

While the general findings shed light on the level of perceived influence on the field by STEM program evaluation, the research team was most interested in the influence of multi-site evaluations conducted in association with the ATE, CETP, LSC, and MSP-RETA programs, their PIs, and related publications.

Recall that 34% of all respondents were able to identify an evaluation by name (n=64). Within this group, only 51 actually named an evaluation. Within this group of 51 named evaluations, 49% (25 of 51) were NSF evaluations but not related to the four programs targeted in this study; 12% (6 of 51) were evaluations associated with other federal non-NSF programs; 29% (15 of 51) were evaluations associated with the targeted programs (i.e., ATE, CETP, LSC, MSP-RETA), and 10% (5 of 51) were associated with locally or privately funded programs.

<u>Publications associated with ATE, CETP, LSC and MSP-RETA</u>

In terms of influential evaluation publications, 30% of all respondents indicated that they were familiar with an influential publication. Although 57 respondents indicated they were familiar with a publication associated with a multi-site STEM education evaluation, only 44 (23% of all respondents) actually named a publication in the follow-up questions. Within this group, 13 of the respondents (6.8% of all respondents) named a publication associated with ATE (n=4), CETP (n=1), LSC (n=4) or MSP (n=4). Almost 16% named publications were associated with other private/local programs (n=7) and 11.4% named publications associated with other federal non-NSF programs (n=5).

When evaluations and publications noted were combined, the research team found that MSP evaluations and publications were noted by 10 respondents, ATE and LSC were each noted by 8 respondents and CETP was noted twice.

Evaluators associated with ATE, CETP, LSC and MSP-RETA

When examining responses regarding known influential evaluators, 59 respondents (30.9% of all respondents) were able to name an influential STEM education evaluator. Of these responses, 29.6% of all respondents named an individual evaluator who was not duplicated in any other responses and the remaining 9.3% are summarized below:

Characteristics of Evaluators Identified as Influential	Responses
An evaluator associated with LSC	7
An evaluator associated with CETP	3
An evaluator associated with American Association for the Advancement of Science	2
An evaluator associated with Inverness Research Associates	2
An evaluator associated with University of Colorado	2
An evaluator associated with Grinnell College	2

This picture changed only slightly when the team looked at the organizational affiliation of all evaluators named. When examining the organizational affiliation of the named evaluators, the percent of named evaluators in the responses who were named more than once increased to 14.4% of all respondents as seen below:

Named Evaluator's Organization	Responses
Horizon Research, Inc.	7
American Association for the Advancement of Science	4
University of Minnesota	3
University of Wisconsin Center for Education Research	2
University of Washington	2
University of Colorado	2
SRI International	2
Michigan State University	2
Inverness Research Associates	2
Grinnell College	2

Focusing further, the research team wanted to explore whether respondents who were involved in one of the four program evaluation projects being examined were more likely than others to name an evaluation, publication, or evaluator from ATE, CETP, LSC, or MSP-RETA. Researchers considered using Chi square analysis to examine the

relationship between involvement and responses regarding influential evaluations, publications, and evaluators. However, given the relatively small size of the sample of respondents involved in each of the four programs, it was not feasible to conduct statistical analysis on the relationship of variables. As such, only frequencies are reported.

Of those respondents who reported they were involved in ATE (n=21), three respondents identified as influential an ATE-affiliated evaluation, publication, or evaluator by name. Two others identified other NSF evaluations, publications, or evaluators as influential to their work.

Of the sample who reported involvement in CETP (n=4), two respondents could identify an influential evaluation. One identified the CETP evaluator as influential, and the other identified another NSF program evaluation, publication, or evaluation.

Likewise, of those who reported involvement in LSC (n=2), one identified an evaluation, publication or evaluator from LSC as influential in their work, and one reported another NSF program's evaluation, publication, or evaluator as influential.

Finally, of those who reported involvement in MSP-RETA evaluation activities (n=8), three identified an MSP evaluation, publication, or evaluator as influential to their work and one noted another NSF program evaluation, publication, or evaluator.

Because sample sizes of those reporting involvement in the four multi-site program evaluation projects was so small, further analysis or modeling of social networks with these findings was not warranted or even possible.

Editor survey findings

None of the three responding editors were able to identify an article published in their own journal prepared by a large scale, multi-site STEM education evaluation.

Only one of the three was able to name an influential evaluation of a large scale, multi-site STEM education evaluation. The evaluation named was the CETP program evaluation, which is one of the four program evaluations targeted in this study.

Two of the three respondents named a total of eight influential STEM education evaluators. Of these eight, two of the STEM education evaluators identified as influential were the PIs of the program evaluations targeted by this study – ATE and LSC.

Two of the three editors identified themselves as non-STEM evaluators, and the third as both a STEM educator/research and a STEM evaluator. This individual was also the only one of the three to report having been personally involved with one of the four targeted, large scale, multi-site program evaluations at the center of this study.

Section 4: Discussion

Given the substantial amount of public funds used to support both education projects and their related evaluation efforts, the research team examined four large multi-site STEM education evaluations to find evidence of influence on the STEM education and evaluation fields. The team defined influence as use by secondary or unintended users. This examination is valuable to the evaluation field because the notion of affecting the field is inherent in the dissemination requirements of the four targeted evaluations, as it is in most NSF grants.

In general, the team found that most respondents were not familiar with any influential evaluations, publications, or evaluators and that even fewer could actually name an influential STEM education evaluation, publication, or evaluator. When examining

subgroups of respondents, those involved with two or more NSF STEM education grants at the time of the survey appeared to have more familiarity with evaluations, publications, and evaluators. The team also found that the respondents' reasons for identifying an evaluation, publication or evaluator as influential were not consistent across the categories. While some similarities were seen in the top selected reason in each category, there were notable differences among the categories. More specifically, the field survey identified only limited influence by the ATE, CETP, LSC and MSP-RETA evaluations on the STEM education field.

The team hypothesized that it would find evidence of each of the four programs being studied at the center of four unique networks with many connections (as indicated by respondents noting evaluations, publications, or evaluators associated with these programs). However, this evidence did not emerge and the team was unable to confirm the presence of a social network of influence for any of the four targeted program evaluation projects. Contrary to the team's initial inclination that the chosen sample might actually bias the results in favor of documenting networks, the results indicated no connection between any of the four STEM education programs and the field being larger than seven connections.

Despite the lack of compelling evidence regarding the influence of these evaluations on the STEM field, the research team does not assert that no influential social network exists in STEM education evaluation. Rather, because the team made specific research design decisions that sacrificed sensitivity for bias reduction (e.g., the survey did not offer fixed response options listing evaluations, publications, or evaluators), it is possible that the methods and sample that were used were not sensitive enough to uncover the presence of such a social network. Some possible alternative explanations are that the networks in the field may be more localized by geography, intellectual specialty, or program association than we were able to capture.

Editor survey discussion

Publication in prominent peer-reviewed journals is a commonly accepted avenue for exerting influence on a field or profession. Hence, the research team looked to the premier peer-reviewed publications in the STEM education and evaluation fields in an attempt to measure the influence of four large-scale, multi-site program evaluations.

As described above, the three editors who responded did not name any articles prepared by a large scale, multi-site STEM education that had been published in their respective journals. It should be noted, however, that the research team was interested in the editors' individual awareness and ability to identify by name any articles, projects, or individuals they thought of as influential in the STEM education and evaluation field.

The fact that no editor named an article does not mean that no influential articles have been published, merely that these editors were unable to recall an article by name. In fact, such articles may have been published in the journal, but not within the current editor's tenure, or that such an article did not appear in this selected subset of peer-reviewed journals during this period. We did not ask the editors to consider the years before they took over the editorship. Moreover, we did not ask them to search their publication archives for an accounting of all the articles produced by large scale, multi-site program evaluations.

Two respondents (one self-identified as a non-STEM evaluator and the other as a STEM educator) were able to name several influential STEM education evaluators. Moreover, this list included the PIs of two of the four targeted program evaluations: ATE and LSC.

Although we elicited responses from only three of the six journal editors, we were pleased to see that some influence of the four program evaluations at the center of this study. The fact that the editors of three major journals in the fields of education and STEM evaluation – who are themselves influential players in those fields – were aware

of and could identify by name without prompting one of the four program evaluations and two of the four evaluation PIs involved is evidence of some influence of large scale, multi-site STEM education evaluations on the field.

Suggestions for future research

There may be a variety of other ways to further investigate the possibility of a network of influence in future research. All of the survey respondents were NSF grant holders and thus represent a group that is more likely to be targeted by the large multi-site program dissemination activities. Yet there was still substantial variety in the size of the projects represented in the sample. The research team was unable to determine from the data collected whether the size of these grants might affect access to dissemination activities or connections to the evaluators or these four programs. So analytically, future research might consider stratifying the sample or other types of samples, like professional societies, by grant size. This method may be more effective in uncovering a network of influence.

Another way that future research may uncover the network of influence of STEM education evaluations on the field might include using other social network analysis methods that differ from the egocentric analysis methods used in the current study. Since egocentric analysis is always about people rather than about groups, it may be more appropriate to examine network connections of STEM educators in naturally occurring groups. By changing the network analysis methods from those that focus on individuals as discrete units of analysis, other analysis methods might focus on how the structure and strength of connections with a variety of professional worlds related to STEM education affects individuals, their relationships, and their access to resources. Likewise, a group of individuals with connections to other social worlds may have access to a wider range of information. This study was unable to examine the influence of relationships in this way.

As part of a wider research effort to determine the effect of project participation on the influence of large scale multi-site program evaluations, this study sought to measure something that has not been measured before. The challenge was to develop a systematic approach to measuring an amorphous concept – influence – on a loosely defined entity – the "field" of STEM education evaluation. To that end, the team employed some of the tools of social network analysis to determine whether the names and titles of evaluation-related people, products, and activities identified by survey respondents as influential would include the names of ATE, CETP, LSC, and MSP-RETA program evaluations, publications, and people. As described above, our ability to use social network analysis techniques was constrained as too few respondents were able to identify projects, products, or people by name. However, despite these limitations, this project has helped us take another step toward sharper definitions of influence in evaluation and has pointed toward some possibilities for future research.

Appendix A: Field influence survey

- Page 1: Intro -

"Influential STEM Program Evaluation" Survey

We appreciate your willingness to take part in this NSF-funded research study on influential large scale, multi-site program evaluations. We are asking for your candid perceptions of influential program evaluations, evaluation publications, and evaluators within the field of STEM education. Your information will enable NSF to enhance its evaluation activities.
The survey takes 5-10 minutes to complete. Please be as detailed as possible in your responses. Your information will be kept confidential and only aggregated results will be reported. By entering your login name and password below, you have consented to let us use the data you provided. Thank you for your time and insights!

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Please think about large scale, multi-site STEM education program evaluations, evaluation publications, and evaluators in terms of their influence on the STEM education field.

Definitions

Large scale, multi-site program evaluation: Any evaluation of a large (nation-wide) program that would include several sites rather than one individual local project. We are interested in program evaluations, as opposed to research studies (e.g., the Trends in International Mathematics and Science Study (TIMSS) or the Schools and Staffing Survey (SASS)) that collect data but are not specifically used to judge the effectiveness of a specific program.

STEM: Science, Technology, Engineering, and Mathematics.

1. Please read the following statements and check yes or no:

I am familiar with one or more influential large scale, multi-site STEM education EVALUATIONS.
_Yes __No

I am familiar with one or more influential PUBLICATIONS produced by or resulting from large scale, multi-site STEM education evaluations.
_Yes __No

I am familiar with one or more influential EVALUATORS who have conducted large scale, multi-site STEM education evaluations.
_Yes __No

(The skip patterns for the above 3 statements should be that if they check yes to any of the below they are taken to the questions regarding that subject. If no to all, they skip to the demographic questions. The skip should be after all three questions were answered.

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2. Based on your experience, please name one influential large scale, multi-site STEM education program evaluation:

Evaluation name:

Evaluator/organization that conducted this evaluation:

For what reason(s) do you think this evaluation has been influential? Check your top 2 choices:
RANDOMIZED SECTIONS (CODE EACH ITEM AS 0 OR 1):
Source of Information about the Evaluation
__This is the only large scale, multi-site STEM education program evaluation of which I am aware.
__I was involved with it.
__My colleagues have frequently used its products.
__I heard about it from people I trust.
__NSF recommended it to me.
__I learned about it in a workshop.
__I learned about it in a reputable publication.
__I was forced to use it.
__Evaluation plans, instruments or reports have been widely disseminated.

Practical Significance
__It evaluated an important program.
__It provided important recommendations.
__It caused/contributed to important changes.
__It has had a far-reaching influence on both policy and practice.
__It produced useful evaluation tools.

Quality of Implementation/Products
__It was high quality work.
__It utilized sound evaluation/research methodology.
__It presented both qualitative and quantitative findings.
__It presented only quantitative findings
__It presented only qualitative findings
__It produced both positive and negative findings.
__Other. Please explain:

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3. Based on your experience, please name one influential PUBLICATION prepared/disseminated by large scale, multi-site STEM education evaluations. (By "publication" we mean an evaluation report, instrument, article, website, etc.)

Publication title:

Publication author:

Type of publication:
__Book
__Brochure
__Evaluation report
__Evaluation instrument
__Journal article
__Webpage
__Other. Please explain: (FOR PILOT ONLY)

For what reason(s) do you think this evaluation publication has been influential? Check your top 2 choices:
RANDOMIZED SECTIONS (CODE EACH ITEM AS 0 OR 1):
Source of Information about the Publication
__This is the only large scale, multi-site STEM education program evaluation publication of which I am aware.
__My colleagues have frequently used it.
__I heard about it from people I trust.
__NSF recommended it to me.
__I learned about it in a workshop.
__I learned about it in a reputable publication.
__It was widely disseminated.

Practical Significance
__It caused/contributed to important changes.
__It considered an important program.
__It has had a far-reaching influence on both policy and practice.
__It provided important recommendations.
__It presented useful evaluation tools.

Quality of Methodology/Products
__It was high quality work.
__It presented sound evaluation/research methodology.
__It presented both qualitative and quantitative findings.
__It presented only quantitative findings
__It presented only qualitative findings
__It presented both positive and negative findings.

__Other. Please explain:

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4. Based on your experience, please name one influential evaluator in the field of STEM education. Please provide that person's full name and organizational affiliation.
Note: Influential evaluators may be affiliated with any type of organization or may be independent or unaffiliated experts.

First name:

Last name:

Affiliation:

This evaluator has been influential because he/she (check your top 2 choices):
RANDOMIZED SECTIONS (CODE EACH ITEM AS 0 OR 1):
Source of Information about the Evaluator
__Is the only evaluator in the field of STEM education of which I am aware.
__Is a personal acquaintance of mine.
__Has worked with me in the past.
__Was recommended by people I trust.
__Was recommended by NSF.
__Was written about in a reputable publication.

Practical Significance
__Evaluated an important program (s).
__Contributed to important changes.
__Contributed significantly to the knowledge base of evaluation.
__Provided important recommendations.
__Helped to strengthen STEM education.
__Mentored many other evaluators.
__Has been a proponent of building evaluation capacity.
__Elevated the discussion about importance of program evaluation.
__Disseminated his/her work (plans, instruments, reports, etc.) widely.

Competencies/Strengths
__Is a high caliber STEM education evaluator.
__Has had considerable experience in STEM evaluation.
__Has demonstrated a deep understanding of STEM education.
__Has utilized sound evaluation/research methodology.
__Other. Please explain:

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Demographics

5. Which of the following statements best describes your experience with NSF grants?
This is my first time as PI of an NSF grant.
I have been a PI on 2-5 NSF grants.
I have been a PI on more than 5 NSF grants.

6. I am a member of (check all that apply):
American Association for the Advancement of Science (AAAS)
American Evaluation Association (AEA)
American Educational Research Association (AERA)
National Association of Research in Science Teaching (NARST)
National Council of Teachers of Mathematics (NCTM)
National Science Teachers Association (NSTA)

7. Which of the following best describes your background?
Scientist/engineer/mathematician
STEM educator/researcher
Non-STEM educator/researcher
STEM evaluator
Non-STEM evaluator
Other:

8. What is your gender?
Male
Female

9. I was personally involved with (check all that apply): (CODE SEPARATELY AS 0-1)
The national evaluation of the Advanced Technological Education (ATE) Program conducted by Western Michigan University
The national evaluation of the Collaboratives for Excellence in Teacher Preparation (CETP) Program conducted by the University of Minnesota
The national evaluation of the Local Systemic Change through Teacher Enhancement (LSC) Program conducted by Horizon Research, Inc.
The national evaluation and technical assistance provided by Utah State University for the Math & Science Partnership (MSP) Program
Another NSF-funded multi-site evaluation: (Space to add name)

Is there anything else you would like to tell us with regard to the survey questions?: (Space to add text)

THANK YOU!